ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 2 SET D **WINTER SEMESTER, 2024-2025**

DURATION: 25 MINUTES **FULL MARKS: 20**

MATH 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID: _____

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1. Given, 2 + 2 (CO2) (PO1)

$$S = \{\begin{bmatrix} x^4 \\ y^{-2} \end{bmatrix} \mid \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2\}$$

This set represents all vectors that are scaled by a factor of 2 from the original vector space $\mathbb{R}^2$. **Is S a vector space?** If yes, justify your answer. If no, provide a counterexample to demonstrate why it is not.

2. 16 (CO2) (PO1)

Given,

$$A = \begin{bmatrix} 1 & -1 & 1 & 0 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 2 & -2 & 2 & 1 & 5 \\ 1 & 0 & 3 & 1 & 3 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

Answer the following questions (After reducing it to its Reduced Row Echelon Form (RREF)): (2×2)

(a) What is the dimension of Matrix A $(m \times n)$? (0.5)

(b) What is the rank of Matrix A? (0.5)

(c) Identify the pivot and free columns of Matrix A. (1)

(d) Find out the **solvability condition** for $Ax = b$. (1)

(e) For all the four fundamental subspaces associated with Matrix A (Columnspace $C(A)$, Nullspace $N(A)$, Rowspace $C(A^T)$, Left Nullspace $N(A^T)$), state the following: $(4 \times 2 + 2 \times 0.5)$

- Their dimensions.
- Bases of the subspaces.
- The subspaces they belong to (i.e., $\mathbb{R}^m$ or $\mathbb{R}^n$, or something else).
- Which two pairs are orthogonal/perpendicular to each other and justify by showing the dot product of their basis vectors. (showing one per pair is enough)

	Columnspace $C(A)$	**Nullspace** $N(A)$
Bases		
Dim		
Subsp.		

	Rowspace $C(A^T)$	**Left Nullspace** $N(A^T)$
Bases		
Dim		
Subsp.		